

Smart Gesture Controls

Group Number: 31

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Description:

The aim of this project is to design a camera-based smart controls system that allows users to control media playback using real-time hand gestures. The system will use a standard webcam as its primary input device, and will interpret predefined hand gestures to trigger actions such as pause/play, volume adjustment, next/previous, and more.

The core focus of the project is the software pipeline, implemented in C++ using OpenCV and MediaPipe to perform real-time gesture recognition. Video frames captured from the webcam will be processed to detect and track the user's hand. From this input, relevant features will be used to identify a set of gestures. Each gesture in the set is then mapped to a specific control command within the media playback system, and then executed through the designated operating system's media playback function using keyboard shortcuts.

Features:

- Hotkey / Gesture to activate smart gesture controls
- Real-time webcam video capture
- Mapping gestures to application-level actions (e.g. slide navigation or internal menu control)
- YouTube/Videoplayer Smart Gestures ex.:
 - High five (pause and play)
 - Swipe left/right twice with pointer finger to skip forward/backward
 - Swipe up/down to increase/decrease volume
 - Multi-hand support (left-hand or right-hand recognition)
 - Point & pinch to zoom in/out
- Support Windows and MacOS

Optional Features:

- On-screen visual feedback (gesture labels, bounding boxes, Fps counter)
- Gesture smoothing and cooldown logic to reduce false positives

- Lighting optimization (allow gestures to work in variety of different lighting conditions)
 - lighting recommendations (flash text recommending to increase or decrease light on the subject for better recognition)

Wish-list Features:

- Records gestures for evaluation and debugging
- PDF/Slideshow/Cookbook smart features
- Temporary Virtual Mouse?
- Virtual checkers?

Risks:

- A risk we might encounter includes accuracy levels due to varying lightings, gesture differentiation, positioning of the person and the arm. We can limit the number of gestures implemented to avoid lower accuracy.
- Using software that group members are not used to. This can be a risk, as there is a higher chance of messing up the code if you're not used to coding in C++. A strategy that we can use to mitigate this risk is having the group members who are proficient in C++ to help the group members who are not, and to ask questions when stuck.
- Performance lag due to real time restrictions. We can use FPS monitoring and higher camera resolutions to mitigate these risks.
- Complexity of integrating third-party libraries
We can mitigate this by using stable, well-documented libraries and maintain a fallback implementation with simpler detection logic

Other notes

- Version control handled through git, using task allocation and branching to support development by multiple members

References:

Open CV documentation: <https://docs.opencv.org/4.x/d1/dfb/intro.html>

MediaPipe documentation: <https://mediapipe.readthedocs.io/en/latest/solutions/hands.html>